

# Diagnostic Evaluation Report

## 1. Framework Information

- \*\*Framework:\*\* X-RAG Diagnostic Framework (v1.0)
- \*\*Trace ID:\*\* `f9ba6791-9da9-4621-b994-f115d2dfcb6b`
- \*\*Timestamp:\*\* 2026-09-05T17:12:37.050131Z
- \*\*Status:\*\* COMPLETED

## 2. Executive Summary

- \*\*Question:\*\* Are [2024] 5 S.C.R. 1074 and 2024 INSC 436 the same decision? State what it decided.
- \*\*Answer:\*\* I do not have enough information to answer this.
- \*\*Health Score:\*\* 0.09
- \*\*Primary Issue:\*\* RETRIEVAL\_MISS
- \*\*Claims Supported:\*\* 0 / 1
- \*\*Summary:\*\* Analysis completed.

## 3. Pipeline Overview

| Stage     | Status  | Confidence | Observation                                                                                     |
|-----------|---------|------------|-------------------------------------------------------------------------------------------------|
| ---       | ---     | ---        | ---                                                                                             |
| CORPUS    | UNKNOWN | 0.5        | Corpus relevance is inconclusive.                                                               |
| RETRIEVER | FAIL    | 0.85       | No retrieved chunks provide support, and retrieval scores strongly indicate a lack of useful ev |
| GENERATOR | UNKNOWN | 0.5        | 1 generated claims are unsupported, but retrieval quality is ambiguous.                         |
| GROUNDING | PASS    | 0.9        | No claims are contradicted by retrieved evidence.                                               |

\*Efficiency: 0/6 chunks used (0%)\*

## 4. Evaluation Metrics

- \*\*Retrieved Chunks:\*\* 6
- \*\*Verified Claims:\*\* 1 (Supported: 0, Partial: 0, Unsupported: 1, Contradicted: 0)
- \*\*Grounding Score:\*\* 0.00
- \*\*Evidence Coverage:\*\* 0.17
- \*\*Average Entailment:\*\* 0.09
- \*\*Latencies:\*\* Retrieval (3418.0ms) | Generation (6550.1ms) | Verification (9278.2ms)

## 5. RAGAS Metrics

- \*\*Faithfulness:\*\* 0.0000
- \*\*Answer Relevancy:\*\* 0.4018
- \*\*Context Precision:\*\* 0.0000
- \*\*Context Relevancy:\*\* 0.0000
- \*\*Context Recall:\*\* N/A
- \*\*Answer Similarity:\*\* N/A
- \*\*Answer Correctness:\*\* N/A

## Answer Correctness (Claim Recall)

Not computed for this trace.

## 6. Evidence Analysis

- \*\*[b9283727-52bd-4c40-a6c7-2acb0bcde28c]\*\* (UNSUPPORTED): I do not have enough information to answer this.
- \*Evidence (Chunk 36cfe22b-f2d7-4380-921c-6e3c0f3c7d75):\* (m) "judicial proceeding" includes any proceeding in the course of which evidence is or may be legally taken on oath;
- (n) "local jurisdiction", in relation to a Court or Magistrate, means the local area within which the Court or Magistrate may exercise all or any of its or his powers under this Sanhita and such local area may comprise the whole of the State, or any part of the State, as the State Government may, by notification, specify;
- (o) "non-cognizable offence" means an offence for which, and "non-cognizable case" means a case in which, a police officer has no authority to arrest without warrant;
- (p) "notification" means a notification published in the Official Gazette;
- (q) "offence" means any act or omission made punishable by any law for the time being in force and includes any act in respect of which a complaint may be made under section 20 of the Cattle Trespass Act, 1871;
- (r) "officer in charge of a police station" includes, when the officer in charge of

the police station is absent from the station-house or unable from illness or other cause to perform his duties, the police officer present at the station-house who is next in rank to such officer and is above the rank of constable or, when the State Government so directs, any other police officer so present;

(s) "place" includes a house, building, tent, vehicle and vessel;

(t) "police report" means a report forwarded by a police officer to a Magistrate under sub-section (3) of section 193;

(u) "police station" means any post or place declared generally or specially by the State Government, to be a police station, and includes any local area specified by the State Government in this behalf;

(v) "Public Prosecutor" means any person appointed under section 18, and includes any person acting under the directions of a Public Prosecutor;

(w) "sub-division" means a sub-division of a district;

(x) "summons-case" means a case relating to an offence, and not being a warrant-case;

## ## 7. Root Cause Analysis

- **Primary Cause:** RETRIEVAL\_MISS

- **Confidence:** 0.88

- **Reasoning:**

- Starting root cause analysis for trace f9ba6791-9da9-4621-b994-f115d2dfcb6b.
- Skipping CORPUS: status is UNKNOWN (insufficient evidence).
- Failure at RETRIEVER: RETRIEVAL\_MISS (confidence=0.85). Observation: No retrieved chunks provide support, and retrieval
- Skipping CHUNKING: no observable state in matrix.
- Skipping GENERATOR: status is UNKNOWN (insufficient evidence).
- GROUNDING passed (confidence=0.90).
- Primary root cause: RETRIEVAL\_MISS at RETRIEVER (earliest upstream failure, confidence=0.85).
- Analysis complete. Primary cause: RETRIEVAL\_MISS.

## ## 8. Corrective Action Plan

- **[IMMEDIATE] Increase Retrieval Top-K**

- Description: Increase the number of retrieved documents before providing context to the LLM.
- Expected Improvement: Higher recall; missing context is caught within the expanded window. (Metric: Verification support)

- **[SHORT\_TERM] Implement Hybrid Search (Dense + Sparse)**

- Description: Add BM25 lexical search and combine with semantic vector search using Reciprocal Rank Fusion.
- Expected Improvement: Near-perfect recall for exact-match terminology. (Metric: Retrieval hit rate for exact entity queries)

- **[EXPERIMENTAL] Deploy Cross-Encoder Reranker**

- Description: Pass the Top-N results through a Cross-Encoder to re-score query-document relevance.
- Expected Improvement: Drastically improved precision at Top-1 and Top-3. (Metric: Precision@3 increases by at least 20%)

- **[INFORMATIONAL] Review Retrieval Top-K / Reranker Cutoff**

- Description: Retrieval efficiency is low (0%); consider reducing top-k or tightening the reranker score cutoff.
- Expected Improvement: Reduced prompt size and lower latency/cost with no loss of verified support. (Metric: Chunk utilization)

## ## 9. Evidence Traceability

- **Primary Cause:** RETRIEVAL\_MISS

- **Derived From:** RootCauseAnalysis

- **Pipeline Stage:** RETRIEVAL\_MISS

- **Supporting Claims:** b9283727-52bd-4c40-a6c7-2acb0bcde28c

- **Supporting Chunks:** 36cfe22b-f2d7-4380-921c-6e3c0f3c7d75

## ## 10. Overall Assessment

- **Major Strength:** GROUNDING is healthy (confidence=90%): No claims are contradicted by retrieved evidence.
- **Major Weakness:** RETRIEVAL\_MISS
- **Next Priority:** Increase Retrieval Top-K
- **Recommendation:** Pipeline requires architectural adjustments across multiple stages.

## ## 11. Framework Footer

- **Generated By:** X-RAG Diagnostic Framework

- **Artifact Version:** 1.0

---

## ## APPENDIX

### Artifacts Generated

#### ✓ RAGTrace

artifacts/rag\_traces/f9ba6791-9da9-4621-b994-f115d2dfcb6b.json

#### ✓ ClaimSet

artifacts/claim\_sets/f9ba6791-9da9-4621-b994-f115d2dfcb6b.json

#### ✓ VerificationResults

artifacts/verification/f9ba6791-9da9-4621-b994-f115d2dfcb6b.json

#### ✓ PipelineStateMatrix

artifacts/pipeline\_state\_matrix/f9ba6791-9da9-4621-b994-f115d2dfcb6b.json

#### ✓ RootCauseAnalysis

artifacts/root\_cause\_analysis/f9ba6791-9da9-4621-b994-f115d2dfcb6b.json

#### ✓ CorrectiveActionPlan

artifacts/corrective\_action\_plan/f9ba6791-9da9-4621-b994-f115d2dfcb6b.json

#### ✓ DiagnosticEvaluationReport

artifacts/reports/f9ba6791-9da9-4621-b994-f115d2dfcb6b.json